Worksheet # 6 Part B Upload solutions
Physics II

Name: _____ Date: _____

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<u>Upload the solution to questions 11 through 15 below</u>

11) For a beam of green light with a wavelength of 530 nm determine

A) the frequency of the light in vacuum.

B) the energy of the green light.

C) Determine the number of green photons in 5 J of green light.

12) An electromagnetic wave is propagating along the positive x direction at a time t and has an electric field given by

$$\vec{E} = (-0.082\ V/m)\hat{y}$$

Determine the magnitude and direction of the magnetic field at time t.

13) A light bulb emits 5 W of optical energy uniformly in all directions.

A) Calculate the intensity of the light at 2 m from the light bulb.

B) Determine the rms electric field strength and magnetic field strength at 2 m.

C) Deterine the distance from the bulb if the electric field is measured to be 2 N/C?

14) An oven heating element is in the shape of a disk (r = 5 cm), is at a temperature of 2,000 °C and is in a room with a temperature of 20 °C. The heating element has an emissivity of 0.90.

A) Determine the net power emitted by the heating element.

B) Determine the peak wavelength emitted by the heating element.

C) The element is turned up so that emits a peak wavelength of λ_{peak} = 678.2 nm, determine the temperature of the heating element.

15) Un-polarized light having an intensity of I_o = 500 W/m^2 is incident on three polarizers, the transmission axis of the first polarizer (P_1) is vertical, the second polarizer's (P_2) transmission axis is rotated by 30º from the vertical and the third polarizer's (P_3) transmission axis is rotated in the same direction as P_2 but 80º from the vertical. What is the intensity of the light exiting the third polarizer (I_3)?

